SIMATS ENGINEERING
ASSESSMENT TOOL 1-APPLICATION- BASED QUESTIONS

COURSE NAME: Cryptography and Network Security
COURSE CODE: CSA51

COURSE OUTCOME COVERED

CO1: Apply classical encryption techniques using symmetric and asymmetric ciphers to encrypt and decrypt data for the ensuring data security. (BL3, BL4)

Assessment Tool Weightage: 40%

QUESTIONS: 5 **Total Marks:100**

Annexure A

APPLICATION- BASED QUESTIONS

Code of Conduct:

I K.Venkatesh(192425290)_ certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:



Annexure A

APPLICATION- BASED QUESTIONS

Scenario: Event Communication Security A student organization is preparing to share confidential event details with its core members using a simple classical encryption method. During testing, a message "MEET AT NOON" is transformed into an encoded version using a fixed shift pattern.
Reconstruct how the encrypted message might appear if a shift of 4 positions is applied.
Explain how an authorized member, aware of the encoding rule, would retrieve the original message.

Answer:

This scenario uses a **Caesar Cipher**, where each letter is shifted by a fixed number of positions in the alphabet.

Encryption using a shift of 4

Original message:

MEET AT NOON

Shift each letter **4 positions forward**:

- M → Q
- E → I
- E → I
- T → X
- A → E
- T → X
- N → R
- O → S
- O → S
- N → R

Therefore, the encrypted message is:

QIIX EX RSSR

Decryption

An authorized member who knows the shift value (**4**) shifts every encrypted letter **4 positions backward**:

QIIX EX RSSR → MEET AT NOON

For example:

- Q → M
- I → E
- X → T
- R → N
- S → O

Thus, the member can recover the original confidential message by applying the **reverse shift of 4 positions**.

Scenario: Legacy Encryption in Corporate Communication A legacy system in a company still uses a transposition-based encryption method with three levels of arrangement. A received message appears as: "WECRLTEERDSOEEFEAOCAIVDEN".

a. Determine the most probable original message by reversing the encryption process.
b. Critically evaluate why such a method is vulnerable in modern communication systems.
c. Propose one practical enhancement to improve the confidentiality of messages in this scenario.

Answer:

Scenario: Legacy Encryption in Corporate Communication

The given ciphertext is:

WECRLTEERDSOEEFEAOCAIVDEN

This resembles a **3-level Rail Fence (transposition) cipher**, where the letters are rearranged rather than replaced.

a. Original Message

By reversing the 3-level transposition arrangement, the most probable original message is:

WE ARE DISCOVERED FLEE AT ONCE

Without spaces:

WEAREDISCOVEREDFLEEATONCE

The encryption only changes the **position of the letters**, while the letters themselves remain unchanged.

b. Why the Method Is Vulnerable

A simple transposition cipher is vulnerable in modern communication because:

1. It uses a **fixed and simple arrangement**, making it easy to recognize.
2. It provides **no strong mathematical security**.
3. An attacker can try different numbers of levels and quickly recover the message.
4. The original letters are preserved, allowing **frequency and pattern analysis**.

Therefore, a basic transposition cipher is unsuitable for protecting confidential corporate communication today.

c. Practical Enhancement

A practical improvement is to **replace the legacy transposition cipher with a modern authenticated encryption algorithm such as AES-GCM**.

AES-GCM provides:

- Strong encryption using a secret key.
- Protection against unauthorized reading.
- Integrity/authentication to detect message modification.
- Much greater resistance to brute-force and cryptanalytic attacks.

Conclusion: The original message is **"WE ARE DISCOVERED FLEE AT ONCE."** While the 3-level transposition method may have been useful historically, it should be replaced with a modern authenticated encryption scheme such as **AES-GCM** for corporate communication.

RUBRICS FOR CALCULATION:

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Understanding of Concepts	25	Demonstrates thorough understanding and effectively integrates multiple concepts/technologies	Good understanding with proper integration of most concepts	Basic understanding g; limited or partial integration	Poor understanding g; unable to integrate concepts
Experiment al Design	30	Well-planned, systematic implementation with optimal solution	Correct implementation with minor issues	Basic implementation on with errors or inefficiencies	Incorrect or incomplete implementation
Use of Tools	20	Effective and advanced use of tools/technologies with proper configuration	Appropriate use of tools with correct execution	Limited or inconsistent use of tools	Incorrect or minimal use of tools
Analysis of Results	15	Insightful analysis with accurate interpretation and validation of results	Good analysis with reasonable interpretation	Basic analysis with limited insights	Poor or incorrect analysis of results
Documentation	10	Clear, well-structured documentation with diagrams, code, and results	Organized documentation with necessary details	Basic documentation with missing details	Poor or incomplete documentation